

Takumi **Shinohara**, Ph.D.

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Employment

- May 2025 – present **Postdoctoral Researcher, KTH Royal Institute of Technology**, Sweden
Research on learning and control of networked controlled systems with Prof. Karl Henrik Johansson.
- Oct. 2024 – Mar. 2025 **Visiting Researcher, Keio University**, Japan
Research on secure control systems with Prof. Toru Namerikawa.
- Apr. 2018 – Apr. 2025 **Consultant, Mitsubishi Research Institute, Inc.**, Japan
Research, study, and consult on the cybersecurity policy for the Japanese Government (e.g., METI, MIC, and NISC) and consult private companies with cybersecurity issues.

Education

- Sep. 2021 – Sep. 2024 **Ph.D. in Engineering** of Keio University, Japan
Thesis Title: *Secure state estimation under sensor attacks*
Advisor: Prof. Toru Namerikawa
- Apr. 2016 – Mar. 2018 **Master in Engineering** of Keio University, Japan
Thesis Title: *Zero-stealthy attacks in cyber-physical systems and secure state estimation in adversarial environments*
Advisor: Prof. Toru Namerikawa
- Apr. 2012 – Mar. 2016 **Bachelor in Engineering** of Keio University, Japan
Thesis Title: *SLAM problem for UAV with considering computational load and unordinary observations*
Advisor: Prof. Toru Namerikawa

List of Publications

I. Journal Articles

- [J1] [Takumi Shinohara](#) and Toru Namerikawa, "Relationship between the number of agents and sparse observability index," *IEEE Open Journal of Control Systems*, 2025. (accepted for publication)
- [J2] [Takumi Shinohara](#) and Toru Namerikawa, "Optimal security investment problem for secure state estimation on cyber-physical systems," *IEEE Transactions on Automatic Control*, vol. 70, no. 2, pp. 1244–1251, 2025.
- [J3] [Takumi Shinohara](#) and Toru Namerikawa, "Optimal resilient sensor placement problem for secure state estimation," *Automatica*, vol. 160, 111454, 2024.
- [J4] [Takumi Shinohara](#) and Toru Namerikawa, "Distributed secure state estimation with a priori sparsity information," *IET Control Theory & Applications*, vol. 16, no. 11, pp. 1086–1097, 2022.
- [J5] [Takumi Shinohara](#), Toru Namerikawa, and Zhihua Qu, "Resilient reinforcement in secure state estimation against sensor attacks with a priori information," *IEEE Transactions on Automatic Control*, vol. 64, no. 12, pp. 5024–5038, 2019.
- [J6] [Takumi Shinohara](#) and Toru Namerikawa, "Reach set-based secure state estimation against sensor attacks with interval hull approximation," *SICE Journal of Control, Measurement, and System Integration*, vol. 11, no. 5, pp. 399–408, 2018.
- [J7] [Takumi Shinohara](#) and Toru Namerikawa, "Perfect stealthy attacks in cyber-physical systems," *Transactions of the Society of Instrument and Control Engineers*, vol. 54, no. 3, pp. 309–319, 2018. (in Japanese)
- [J8] [Takumi Shinohara](#) and Toru Namerikawa, "On the vulnerabilities due to manipulative zero-stealthy attacks in cyber-physical systems," *SICE Journal of Control, Measurement, and System Integration*, vol. 10, no. 6, pp. 563–570, 2017.
- [J9] Takashi Irita, [Takumi Shinohara](#) and Toru Namerikawa, "Detection of replay attack on smart grid with code signal and bargaining game," *Transactions of the Society of Instrument and Control Engineers*, vol. 52, no. 9, pp. 498–506, 2016. (in Japanese)

II. Referred Conference Papers

- [C1] [Takumi Shinohara](#) and Toru Namerikawa, "Secure state estimation, attack detection, and safe control against sensor attacks with side initial state information." (under review)
- [C2] [Takumi Shinohara](#) and Toru Namerikawa, "Security measure implementation for distributed state estimation," in *Proc. 5th IFAC Workshop on Cyber-Physical Human Systems*, Antalya, Türkiye, 2024.
- [C3] [Takumi Shinohara](#) and Toru Namerikawa, "Secure state estimation for multi-agent systems: On the relationship between the number of agents and system resilience," in *Proc. 2023 American Control Conference*, San Diego, CA, 2023, pp. 1006–1011.
- [C4] [Takumi Shinohara](#) and Toru Namerikawa, "Reach set-based attack resilient state estimation against omniscient adversaries," in *Proc. 2018 American Control Conference*, Milwaukee, WI, 2018, pp.

5813–5818.

- [C5] Takumi Shinohara and Toru Namerikawa, “Manipulative zero-stealthy attacks in cyber-physical systems: Existence space of feasible attack objectives,” in *Proc. 1st IEEE Conference on Control Technology and Applications*, Kohala Coast, HI, 2017, pp. 1123–1128.
- [C6] Takumi Shinohara and Toru Namerikawa, “SLAM for a small UAV with compensation for unordinary observations and convergence analysis,” in *Proc. 2016 55th Annual Conference of the Society of Instrument and Control Engineers of Japan (SICE)*, Tsukuba, Japan, 2016, pp. 1252–1257.

Honors and Awards

- Wallenberg Postdoctoral Scholarship, Knut and Alice Wallenberg Foundation, 2025–present.
- Fujiwara award (Best Ph.D. student award), Keio University, 2025.
- Best presentation award, 65th Japan Joint Automatic Control Conference, 2022.
- Best presentation award, 59th Japan Joint Automatic Control Conference, 2016.

Review Activities

- *Journals*
 - IEEE Transactions on Automatic Control
 - Automatica
 - IEEE Transactions on Control of Network Systems
 - IEEE Transactions on Cybernetics
 - IEEE Journal on Selected Areas in Communications
- *Conferences*
 - IEEE Conference on Decision and Control
 - American Control Conference
 - IEEE Vehicular Technology Conference